UNSW SYDNEY

UNSW Business School
School of Risk and Actuarial Studies

ACTL3142/5110: Statistical Machine Learning for Risk and Actuarial Applications

Assessment Task	Assignment Part 2
Due date	Friday 7th August 2026, 11:55am
Weight (%)	18%
Task description	Individual Assignment
Submission Method	Moodle Turnitin
Feedback mechanisms	Breakdown of marks per question, individualised comments & global feedback with "common mistakes".
Links to Unit Learning Outcomes	CLO1, CLO2, CLO3, CLO5

INTEGRITY MATTERS
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Contents

1. Context

In this assignment, you will continue analysing the airbnb-Sydney.csv data set, which should be familiar to you because of previous assessments. The data is found under Ed Resources and contains information about ∼8,000 Airbnb listings in Sydney, Australia. In this assignment, you have two main tasks:

- A "classification" task, where we are interested in **understanding** the factors that explain why an Airbnb listing has an especially high "user score".
- A "regression" task, where we are interested in **predicting** the price_per_room of a given Airbnb listing (as function of relevant predictors).

In completing these tasks, assume that you are an analyst working for a private investor who is looking to buy properties in Sydney and potentially list them on Airbnb to generate rental income. The investor is also interested in understanding how to optimise the "user score" of Airbnb listings she would aquire, where possible.

You should answer all the questions in the next section in a well organised report, in PDF format (see also Section 3: Directives and Format Requirements).

2. Your Tasks

1. (5pt) We start with the classification task, for which the **response variable** is "rating_q1". As stated in the Data Dictionary, this variable is binary and defined as equal to:
   - **1** if a listing's review_scores_rating is at least as high as its neighbourhood 75%-quantile of review_scores_rating;
   - **0** otherwise.

   Your task is to create the best classification model that you can to classify an Airbnb listing as having either "rating_q1=1" or "rating_q1=0". You are free to use any model you think suitable, and any of the variables in the data set as predictors **except** the 7 variables that start with review_scores_. Note this is forbidden because it would constitute data leakage. Indeed, in a realistic setting, we would try to predict "rating_q1" for a **new listing** for which we do *not* have access to any "user ratings".

   As in Assignment Part 1, you can transform the variables and/or create new ones from the existing ones and/or include interactions between variables (but you still **cannot** use the 7 variables that start with `review_scores_`). You also can't import new variables from external sources.

(a) Present a summary of your model (containing all the information needed to interpret your model).

(b) Carefully explain and justify what makes it the best model you could create.

(c) Explain how good "overall" your classification model is, and whether it has any important issues/flaws.

Where relevant, don't neglect to justify your answers with appropriate diagnostic tools.

Note: as in Assignment Part 1, you are in charge of choosing what makes a model "the best", but you need to clearly explain your rationale. In addition, for this task, while the predictive capacity of your model is important, **interpretability** is *also* important when choosing the best model (see also Question 2).

2. (2pt) Describe the main findings from your model. In particular, what insights do your results provide to a Sydney investor owning Airbnb listings which **do not** have "`rating_q1=1`" (and who wishes to remedy that)?

3. (5pt) We now move on to the "regression" task. Denote "Y" the `price_per_room` of an Airbnb listing[1]. Create the best model that you can to **predict** Y via the predictors in the data set.

   Here, the focus is on **prediction rather than interpretation**. That is, your goal is to minimise the **test MSE** of your model. You are free to use any model you think suitable, and any of the variables in the data set as predictors. Again, you can transform the variables and/or create new ones from the existing ones and/or include interactions between variables (but you are not allowed to import new variables from external sources).

   (a) Present a summary of your model (containing all the information needed to interpret your model).

   (b) Carefully explain and justify what makes it the best model you could create.

   (c) Comment on how good you think your model is at predicting `price_per_room`.

4. (1.5pt) If this was a real problem that you were faced with and you had a lot more time and resources, what *realistic* (but still *ambitious* and possibly *difficult*) additional steps could you take to further improve your "best predictive model" from Question 3? Please justify these recommendations.

5. (2.5pt) Use your predictive model from Question 3 to **predict** the price of the Airbnb listings in the "test set", i.e., in the file `AirbnbTest.xlsx` on Ed. **Very Important:**

   - You must submit a file with your predictions on Turnitin (along with your report). This file must be identical to `AirbnbTest.xlsx`, only with your **predictions added in the last column** "`AY`". The name of that column (cell AY1) must be your **zID**: the lower-case "z" followed by 7 numbers, e.g., z1234567

[1] Recall that `price_per_room` represents the daily price of an Airbnb listing divided by the total number of bedrooms available to the customer booking the Airbnb. If the number of bedrooms is "0", then `price_per_room` is just equal to "daily price".

- Your predictions should be formatted as **numbers** representing the dollar price **without dollar signs**.
- **Do not add or remove rows to this .xlsx file (and do not shuffle the rows' order).**
- Watch out that the format of the "original" dataset is **.csv**, but the format of the test set is **.xlsx**. The format of your submitted predictions' file must be **.xlsx**.

As we will automate the grading of Question 5, failure to respect the above directives will unfortunately result in a 0 for this question. Note that if you leave a prediction empty for a given row, we will assume your prediction is "0".

3. Directives and Format Requirements

All students must submit an assignment of their **individual own work**. Students must comply with the following directives and format requirements. Penalties will be applied to any breach of such directives (and we will not grade any work that exceeds the page limit).

- You must submit your assignment via Turnitin (on **Moodle**), and you have **3 distinct files** to submit:
  - a **.pdf** file: this is your report.
  - a **.xlsx** file with your predictions (see Question 5 for exact instructions).
  - a file with all your computer code.
- Only **typed assignments (not handwritten)** will be accepted.
- The `.pdf` file should include a title page with: student name, student ID.
- The **main body** of your `.pdf` report (excluding title page, appendices, bibliography) must not exceed **4 pages**. It must contain everything that is asked in the questions (e.g., results, explanations, plots, diagnostics, etc.), but should not include your computer code. For Q1 and Q3, you are permitted to place your "model summaries" in an Appendix. Where relevant, please show your reasoning and thought process as clearly as possible in your answers.
- You must **number** your answers, in the same way the questions are numbered.
- Your report must include an appendix titled "Generative AI usage" explaining what you used AI for in this assignment. See Section 4.2 for more details.
- All computer code necessary to produce your results **must ALSO be placed in an Appendix** to your report. There is no page limit for this Appendix.
- This code within the appendix must be made of **text** (not images), that we can copy-paste.

- Your code in the **code file** must run “as is” (and produce exactly the results in your assignment). If we cannot run your code, you will lose ALL marks attached to results this code is meant to produce.
- The page format to use is A4 (this is the standard Australian format).
- The minimum font size must be an equivalent of ‘Times New Roman’ size 11.
- The minimum line spacing to use is 1.15.
- The margins must not be narrower than the ‘narrow’ option in Word (which is 0.5 inches every side).
- Your code should also include the relevant question numbers (as comments, e.g., `#Q1, #Q2, etc.`).

4. Marking Guidelines

4.1 Marking Criteria

Individual questions are allocated a fixed number of marks. Questions 1 to 4 are evaluated using Criteria C1 and C2 as defined below (the descriptions provided correspond to a “HD mark”). Question 5 is assessed only via criteria C4. The overall “Presentation and Professionalism” of your report also counts for 2/18 of the mark, and will be assessed under criteria C3 below. Lastly, we will deduct marks for any breach of the instructions & requirements detailed in Section 3.

- **C1: Methods and Results.** Chooses and correctly applies appropriate methods to derive relevant results. Provides comprehensive and compelling reasoning to justify all components of these methods. Where relevant, uses correct diagnostic tools to assess the appropriateness and/or performance of the methods used. Demonstrates that many different models/methods have been considered, and that the best ones have been selected based on sensible criteria. Calculations, computer codes and other workings produce results which are correct and answer the questions asked. Does not produce any incorrect, irrelevant or superfluous results.
- **C2: Analysis and Communication.** Performs insightful and accurate analysis of the results obtained. Interprets findings correctly and precisely. Shows a great depth of understanding and critical thinking when interpreting/analysing findings. Accurately identifies and explains the assumptions and limitations of the methods used. Explains the relevant nuances and caveats of the results obtained, where necessary. Expresses all ideas clearly and with focus, with no irrelevant or repetitive material. Develops ideas in a logical and coherent manner. Presents opinions and arguments logically and persuasively. Uses fluent, precise and concise language which is appropriate for the intended audience, context and purpose.

- **C3: Presentation and Professionalism.** Presents all findings very professionally. Shows evidence of thorough editing (e.g., no typos, grammar, spelling or punctuation errors). Chooses style(s) of presentation appropriate for the specific business/academic context. Formats and presents results clearly and neatly. Displays figures and tables in a readable and aesthetic format. Chooses appropriate visualisation and/or data summary methods that convey all (and only) the appropriate information.
- **C4: Accuracy of Predictions.** For a model whose stated objective is to produce good predictions, the top 20% of students' assignments (based on the MSE of their test set predictions) will receive full marks. Any student whose test MSE is as bad or worse than the most naive model (i.e., a model predicting $\bar{y}$ for every observation) will receive 0 marks. All other submissions will receive marks in proportion to the 20th percentile MSE.

4.2 Plagiarism Awareness and AI Use

This is an individual assignment. The material each student submits must be their own individual work. Students should make sure they understand what plagiarism is. Note that plagiarism rules also apply to your computer code, which must be created from your own device, and developed by you. The best strategy to avoid any problem is to never share pieces of your work with other students.

You are allowed to use Generative Artificial Intelligence (AI) to help you with editing, planning, idea generation, or coding. However, you must include an Appendix in your report titled "Generative AI usage" explaining what you used AI for and, if applicable, outlining what prompts you used. If you did not use Generative AI, you must still write in this Appendix that generative AI was not used. Copying and pasting computer code, sentences, or paragraphs from AI tools without proper citation is considered plagiarism (in the same way it would be considered as plagiarism if you copy-pasted sentences from your friend's assignment, or any other source). When referencing AI, you must follow UNSW's official guideline as provided here: https://www.student.unsw.edu.au/ai-referencing.

4.3 Late Submission

Any late submission will incur a penalty of 5% per day or part thereof (including weekends) from the due date and time. An assessment will not be accepted after 5 days (120 hours) of the original deadline unless special consideration has been approved. In the case of an approved Equitable Learning Plan (ELP) provision, special consideration or short extension, the late penalty applies from the date of approved time extension. After five days from the extended deadline, the assessment cannot be submitted.

4.4 Short Extension

"Short Extension" is a new process that allows you to apply for an extended deadline on your assessment without the need to provide supporting documentation, offering immediate approval during brief, life-disrupting events. Requests are automatically approved once submitted. For this assignment, a Short Extension of **2 days** is available. Details on how to submit a Short Extesion request (or a “Special Consideration” request) are found here: https://www.student.unsw.edu.au/special-consideration. Note you must do this **before** the official due date of the assignment.

5. Students’ Questions

5.1 Answering Students’ Questions on Ed

Questions about the assignment must be posted on Ed Discussion, under the category “Assessments”. We do not plan to give out many additional instructions or hints, but **if** we do, it is only fair that everyone will have access to the response. Please note that there can be a delay of a few days between a question and its answer. In particular, we do not plan to answer questions posted less than 48h before the official deadline. Hence, please get your questions in by Wednesday 11:55am of Week 10.

Lastly, before posting a question, please have a look to see if your question has already been answered in an existing thread.